

MAT464 Lecture Notes

Arky!! :3c

'26 Fall Semester

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Day 1: Recap of Differential Geometry (Sep. 8, 2026)

Tests will involve slightly more advanced problems than quizzes; there will be no homework, but you are suggested to do the exercises in the textbook.

We start with smooth manifolds. Let X be a topological space. For all points $p \in X$, there exists an (open) neighborhood $U_\alpha \ni p$, with a chart $\varphi_\alpha: U_\alpha \rightarrow V_\alpha \subset \mathbb{R}^n$ between open sets. Denote χ_α to be the inverse of φ_α , and consider another chart $\varphi_\beta: U_\beta \rightarrow V_\beta$ with $p \in U_\beta$ and $\chi_\beta := \varphi_\beta^{-1}$. Then we have two diffeomorphisms,

$$\begin{aligned}\varphi_{\alpha\beta} &:= \chi_\beta \circ \chi_\alpha^{-1}: U_\alpha \rightarrow U_\beta, \\ \varphi_{\beta\alpha} &:= \chi_\alpha \circ \chi_\beta^{-1}: U_\beta \rightarrow U_\alpha.\end{aligned}$$

Here, we will also assume that X is Hausdorff and second-countable. As an example, consider $f: \mathbb{R}^{n+1} \rightarrow \mathbb{R}$, where c is a regular value in $\mathbb{R}$. Then $\{f = c\}$ is a n -manifold in $\mathbb{R}^{n+1}$. For all $x \in \{f = c\} = M$, we see that

$$\nabla f = \left(\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_{n+1}}(x) \right)$$

has maximal rank 1 if and only if $\nabla f(x) \neq 0$, which is equivalent to there being some index i such that

$$\frac{\partial f}{\partial x_i}(x) \neq 0.$$

He then said something incomprehensible, but this is mostly covering basics of differential geometry, which is why (I think) he is going so fast. We move onto his next example. Consider $f: \mathbb{R}^{n+1} \rightarrow \mathbb{R}$, where $f(x) = x_1^2 + \dots + x_{n+1}^2$. We then claim that $c = 1$ is a regular value. Indeed,

$$\nabla f(x) = (2x_1, \dots, 2x_{n+1}),$$

where we see that, if the sum of the x_i^2 terms are equal to 1, then some of the x_i 's are nonzero. If there is some nonzero x_i , then we may solve for x_i by writing

$$x_i = \sqrt{1 - x_1^2 - \dots - \widehat{x_i^2} - x_n^2}.$$

We now move onto real projective space, denoted $\mathbb{RP}^n$. $\mathbb{RP}^n$ consists of the lines through 0 in $\mathbb{R}^{n+1}$, for which the homogeneous coordinates are given by $[x_0 : x_1 : \dots : x_n]$. The manifold structure on $\mathbb{RP}^n$ is given by the following charts

$$X^i: U_i \rightarrow \mathbb{R}^n, \quad [x_0 : x_1 : \dots : x_n] \mapsto \left(\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i} \right),$$

where $U_i = \{[x_0 : \dots : x_n] \mid x_i \neq 0\}$. We see that these charts are well defined because $\mathbb{RP}^n$ identifies points in $\mathbb{R}^{n+1}$ together if they are a scalar multiple apart, but said scalar is cancelled out under the maps X^i . It is left as an exercise to check that the transition maps are smooth between each U_i and U_j .

Similarly, the complex projective space $\mathbb{CP}^n$ consists of complex lines through 0; it is a $2n$ -manifold in $\mathbb{C}^{n+1} \cong \mathbb{R}^{2n+2}$.

Let M^n and N^m be smooth manifolds. A map $f: M^n \rightarrow N^m$ is smooth if it is smooth in charts.

A tangent vector v at $p \in M$ is an equivalence class of smooth curves through p . Specifically, consider $\gamma: (-1, 1) \rightarrow M$, with $\gamma(0) = p$. We say two curves γ_1 and γ_2 are equivalent if their tangent vectors at 0 are equal in charts to $\mathbb{R}^n$, i.e., consider $\hat{\gamma}_1$ and $\hat{\gamma}_2$ as $X_\alpha \circ \gamma_1$ and $X_\alpha \circ \gamma_2$ respectively; then equivalence is asking for $\hat{\gamma}_1(0) = \hat{\gamma}_2(0)$.¹

¹it is at this point that i realized the χ 's are actually just X 's. how do i read his handwriting come on

We may write

$$X_\beta \circ \gamma_1 = (X_\beta \circ X_\alpha^{-1}) \circ (X_\alpha \circ \gamma_1) = \gamma_{\beta\alpha} \circ (X_\alpha \circ \gamma_1),$$

where, by the chain rule, we get

$$(X_\alpha \circ \gamma_1)'(0) = d\varphi_{\beta\alpha}((X_\alpha \circ \gamma_1)'(0)).$$

Similarly, we may write a statement for γ_2 .

We now talk about derivations. A derivation at p is a linear map $D: C^\infty(M^n) \rightarrow \mathbb{R}$ (remember that $C^\infty(M^n)$ is the space of smooth functions on M), such that

$$D(fg) = D(f)g(p) + f(p)D(g).$$

Some other properties of derivations are as follows:

$$D(\mathbf{1} \cdot \mathbf{1}) = D(\mathbf{1}) \cdot 1 + 1 \cdot D(\mathbf{1}) = 2D(\mathbf{1}) \implies D(\mathbf{1}) = 0.$$

Similarly, $D(c\mathbf{1}) = cD(\mathbf{1}) = 0$. If $f(p) = g(p) = 0$, then $D(f \cdot g) = 0$. He then wrote down some more things that were basically indecipherable because of shitty handwriting.

Exercise 1.1. This canonical map arising from $D_v: C^\infty(M) \rightarrow \mathbb{R}$, given by $D_v(f) := (f \circ \gamma)'(0)$, gives an equivalence class of curves at p , and is well-defined.

Consider $\hat{f} = f \circ \gamma: (-1, 1) \rightarrow \mathbb{R}$ and $\hat{g} = g \circ \gamma: (-1, 1) \rightarrow \mathbb{R}$. Then the Leibniz rule applies,

$$(\hat{f} \cdot \hat{g})'(0) = \hat{f}'(0) \cdot \hat{g}(0) + \hat{f}(0) \cdot \hat{g}'(0).$$

He then wrote a theorem here about some bijection between derivations and tangent vectors. Since he's really just recapping MAT367, I'll stop recording notes for today. One bit of notation different from what you see in MAT367, though, is that he uses $\text{Der}_p(M)$ to denote the derivations at $p \in M$. Also, $\text{Der}_p(M) = T_p M$.

Day 2: (Sep. 10, 2026)

Quiz 1 will be on Tuesday of next week. Let M^n be a smooth manifold, and let $p \in M$. We have that $T_p M$ is the space of tangent vectors at p . There are two ways of looking at tangent vectors. First, we can think of them as equivalence classes of smooth curves through p , i.e., $\gamma_1, \gamma_2: (-1, 1) \rightarrow M$ with $\gamma_1(0) = \gamma_2(0) = p$. We say that $\gamma_1 \sim \gamma_2$ if $(X \circ \gamma_1)'(0) = (X \circ \gamma_2)'(0)$, where X is the chart from a neighborhood of p to $\mathbb{R}^n$. Second, we may think of it as $T_p M \cong \text{Der}_p(M)$, where $\text{Der}_p(M)$ are the derivations at p , i.e., the linear maps $D: C^\infty(M) \rightarrow \mathbb{R}$ that satisfy

$$D(f \cdot g) = D(f)g(p) + f(p)D(g).$$

The equivalence class of γ , $[\gamma]$, defines a derivation D by the formula

$$D(f) = \left. \frac{d}{dt} (f(\gamma(t))) \right|_{t=0}.$$

$[\gamma] \mapsto D$ is a bijection. $T_p M$ is a vector space, and we can find bases of $T_p M$ from coordinate charts. For X a coordinate chart near p , we have a basis of $T_p M$ given by

$$\left. \frac{d}{dX_1} \right|_p, \dots, \left. \frac{d}{dX_n} \right|_p.$$

This holds for any chart X . Indeed, to make any general derivation $D \in \text{Der}_p(M)$ in this basis, we may simply just write

$$D = \sum_i v_i \left. \frac{d}{dX_i} \right|_p, \quad v_i \in \mathbb{R}.$$

In the special case, let $M = V$ be an open subset of $\mathbb{R}^n$. Then $T_p M \cong \mathbb{R}^n$ canonically if we think about it as equivalence classes of curves. Specifically, $[\gamma_1] = [\gamma_2]$ if $\gamma_1'(0) = \gamma_2'(0)$ (are the usual vectors in $\mathbb{R}^n$). Similarly, let $F: V \rightarrow \mathbb{R}^m$, where $V \subset \mathbb{R}^n$ is open, and $n \geq m$. Then $\langle F = c \rangle$ is a manifold of dimension $n - m$, and for all $x \in \langle F = c \rangle$, we would have that

$$\left(\frac{df_i}{dX_j}(x) \right)$$

is of rank m . Tangent vectors to $\langle F = c \rangle$ can be thought of as vectors in $\mathbb{R}^n$. As another example, let $F: \mathbb{R}^{n+1} \rightarrow \mathbb{R}$, where $F(x_0, \dots, x_n) = \sum_i x_i^2$, and let $c = 1$; then $x_0^2 + \dots + x_n^2 = 1$ is precisely $\langle F = 1 \rangle$, and we have $T_p S^n = p^\perp$, for $p \in \langle F = 1 \rangle = S^n$.

We will now discuss the differential of a smooth map. Let $f: M^m \rightarrow N^n$ be smooth. f induces a linear map

$$df_p: T_p M \rightarrow T_{f(p)} N,$$

where, for $v = [\gamma] \in T_p M$, we have $df_p([v]) = [f \circ \gamma]$. Alternatively, by regarding $v = \gamma'(0)$, we have that $df_p(v) = (f \circ \gamma)'(0)$. If we think about tangent vectors of derivations, where $h \in C^\infty(N)$, then $df_p(D)(h) := D(h \circ f)$. For $D \in \text{Der}_p(M)$, we have that $df_p(D) \in \text{Der}_{f(p)}(N)$.

Let Y be a chart near $f(p)$. Let

$$\bar{w} = df_p(\bar{v}) = \sum_{j=1}^m w_j \left. \frac{d}{dY_j} \right|_{f(p)}.$$

How do we find the coefficients? $w_j = w(y_j) = df_p(v)(y_j) = v(y_j \circ f)$. He then did a bunch of stuff about transport of structure without really any... direction.

We now talk about tangent bundles. The tangent bundle of M^n , a smooth manifold, is TM , where

$$TM = \langle (p, v) \mid p \in M, v \in T_p M \rangle.$$

Let $\pi: TM \rightarrow M$ be the canonical projection. If $X: U \rightarrow V \subset \mathbb{R}^n$ is a coordinate chart near p , then X maps

$$\left(p, \sum_{i=1}^n v_i \frac{d}{dX_i}(p) \right) \rightarrow (X(p), (v_1, \dots, v_n)) \subset V \times \mathbb{R}^n \subset \mathbb{R}^{2n}.$$

He then talked about charts about π . I'm not following anything he's writing honestly.